A functor is a left adjoint exactly when the hom-functors hom_B(F a, b) are representable

The Univalent Foundations Program

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Editorial note. The counted statement is the author's ct:adj-repr (source lines 1010-1016); the proof environment begins on the line immediately after the statement, so the pairing is by adjacency.

Editorial note. The retained context is the author's definition of representability, the theorem that representability is a mere proposition together with its proof, and the two sentences that introduce the criterion, retained because the statement is phrased with representability and because the following corollary re-proves the adjointness criterion from it.

Editorial note. The retained counter values (chapter 9, section 5, theorem offset 7) reproduce the first-edition numbering: the retained definition prints as Definition 9.5.8, the retained theorem as Theorem 9.5.9, and the counted lemma as Lemma 9.5.10.

Editorial note. Two references leave the retained ranges, to the Yoneda corollary of the same section and to the adjointness lemma of an earlier section; both are declared as non-load-bearing source pointers and replace no step of the counted proof.

Editorial note. No retained line refers to a figure, bibliography entry, included file or non-ASCII character, so the package declares no asset and no bibliography.

Editorial scope. One original result of the HoTT Book, counted once: a functor is a left adjoint exactly when the hom-functors hom_B(F a, b) are representable.

Allowed standard prerequisites (editorial).

- Representable functors. A functor F : Set^(A^op) is representable when there merely exists a : A with a natural isomorphism y a = F, where y is the Yoneda embedding.. Clause (ii) of the counted statement is that for each b : B the functor a |-> hom_B(Fa,b) is representable; the definition is retained as context at source lines 991-996.
- Adjunctions by unit, counit and triangle identities. F : A -> B is a left adjoint when there are G : B -> A, eta : 1_A -> GF, eps : FG -> 1_B satisfying the triangle identities (eps F)(F eta) = 1_F and (G eps)(eta G) = 1_G.. Clause (i) of the counted statement is that F is a left adjoint, and the proof reconstructs G, eta and eps from the representing isomorphisms

(source lines 1022-1025) and conversely builds the representing family from them (source lines 1028-1039).

- The Yoneda embedding is an embedding and representability is a mere proposition. By the Yoneda lemma the embedding y : A -> Set^(A^op) is fully faithful, and if A is a category then y_0 is an embedding, so representability of a functor is a mere proposition (book Corollary 9.5.7 and Theorem 9.5.9).. Retained as context at source lines 998-1004, where the author records the mere-propositionality of representability that makes clause (ii) a property rather than structure.

Source reference. Original corollary 9.5.7: sources/hott-categories/pinned-original-source.tex, source lines 982–982. The retained context proof cites the corollary that the Yoneda embedding is an embedding on objects. It is a navigation pointer to the immediately preceding results of the same section: the counted lemma's own author proof is retained line for line and no step of it is replaced by this reference.

Source reference. Original lemma 9.3.2: sources/hott-categories/pinned-original-source.tex, source lines 504–504. The retained sentence announces that the representation criterion will be used to give another proof of the earlier lemma that being a left adjoint is a mere proposition. It is a navigation pointer only; nothing in the counted statement or its retained proof depends on that pointer.

Definition 9.5.8. A functor $F : \mathcal{S}et^{A^{\mathrm{op}}}$ is said to be **representable** if there exists $a : A$ and an isomorphism $\mathbf{y}a \cong F$.

Theorem 9.5.9. *If A is a category, then the type "F is representable" is a mere proposition.*

Proof. By definition "F is representable" is just the fiber of $\mathbf{y}_0$ over F. Since $\mathbf{y}_0$ is an embedding by original corollary 9.5.7, this fiber is a mere proposition. □

In particular, in a category, any two representations of the same functor are equal. We can use this to give a different proof of original lemma 9.3.2. First we give a characterization of adjunctions in terms of representability.

Lemma 9.5.10. *For any precategories A and B and a functor $F : A \to B$, the following types are equivalent.*

(i) F is a left adjoint.
(ii) For each $b : B$, the functor $(a \mapsto \hom_B(Fa, b))$ from A^{op} to $\mathcal{S}et$ is representable.

Proof. An element of the type (ii) consists of a function $G_0 : B_0 \to A_0$ together with, for every $a : A$ and $b : B$ an isomorphism

$$\gamma_{a,b} : \hom_B(Fa, b) \cong \hom_A(a, G_0 b)$$

such that $\gamma_{a,b}(g \circ Ff) = \gamma_{a',b}(g) \circ f$ for $f : \hom_A(a, a')$.

Given this, for $a : A$ we define $\eta_a :\equiv \gamma_{a,Fa}(1_{Fa})$, and for $b : B$ we define $\epsilon_b :\equiv (\gamma_{Gb,b})^{-1}(1_{Gb})$. Now for $g : \hom_B(b, b')$ we define

$$G_{b,b'}(g) :\equiv \gamma_{Gb,b'}(g \circ \epsilon_b)$$

The verifications that G is a functor and η and ϵ are natural transformations satisfying the triangle identities are exactly as in the classical case, and as they are all mere propositions we will not care about their values. Thus, we have a function (ii)→(i).

In the other direction, if F is a left adjoint, we of course have G_0 specified, and we can take $\gamma_{a,b}$ to be the composite

$$\hom_B(Fa, b) \xrightarrow{G_{Fa,b}} \hom_A(GFa, Gb) \xrightarrow{(-\circ\eta_a)} \hom_A(a, Gb).$$

This is clearly natural since η is, and it has an inverse given by

$$\hom_A(a, Gb) \xrightarrow{F_{a,Gb}} \hom_B(Fa, FGb) \xrightarrow{(\epsilon_b\circ -)} \hom_A(Fa, b)$$

(by the triangle identities). Thus we also have (i)→ (ii).

For the composite (ii)→(i)→ (ii), clearly the function G_0 is preserved, so it suffices to check that we get back γ. But the new γ is defined to take $f : \hom_B(Fa, b)$ to

$$\begin{aligned} G(f) \circ \eta_a &\equiv \gamma_{GFa,b}(f \circ \epsilon_{Fa}) \circ \eta_a \\ &= \gamma_{GFa,b}(f \circ \epsilon_{Fa} \circ F\eta_a) \\ &= \gamma_{GFa,b}(f) \end{aligned}$$

so it agrees with the old one.

Finally, for (i)→(ii)→ (i), we certainly get back the functor G on objects. The new $G_{b,b'} : \hom_B(b, b') \to \hom_A(Gb, Gb')$ is defined to take g to

$$\begin{aligned} \gamma_{Gb,b'}(g \circ \epsilon_b) &\equiv G(g \circ \epsilon_b) \circ \eta_{Gb} \\ &= G(g) \circ G\epsilon_b \circ \eta_{Gb} \\ &= G(g) \end{aligned}$$

so it agrees with the old one. The new η_a is defined to be $\gamma_{a,Fa}(1_{Fa}) \equiv G(1_{Fa}) \circ \eta_a$, so it equals the old η_a. And finally, the new ϵ_b is defined to be $(\gamma_{Gb,b})^{-1}(1_{Gb}) \equiv \epsilon_b \circ F(1_{Gb})$, which also equals the old ϵ_b. □